Pavan Kumar Reddy Devulapalle

(469) 666-1760 | pavandevulapalle@gmail.com | LinkedIn

EXPERIENCE

Software Development Engineer, Amazon Web Services, Seattle, WA — Oct 2024 - Nov 2025

– Developed critical modules of org-level **Local Zones opt-in** platform enabling enterprise-wide activation via a **single API call** using **Java**, **Spring Boot, Kafka**, **Redis**, improving onboarding efficiency by **60%+.**
– Built a self-service admin console using **React** and **TypeScript** for org admins to view eligibility, enable zones, and track rollout state across accounts, reducing support-driven enablements.
– Built an event-driven control plane using **Kafka** for async fan-out, retry, auditability of multi-account operation.
– Added distributed **rate limiting** and per-tenant **quotas** to protect downstream dependencies.
– Standardized retry policies with exponential backoff, **jitter** to reduce correlated retries and improve success rates during regional dependency degradation.
– Used **Redis** (cache + lightweight locks) to accelerate read-heavy org state lookups and prevent duplicate work.
– Containerized microservices with **Docker** on **Kubernetes** (EKS); managed releases via Helm for deployments.
– Led the "Provisioning at ODM" initiative for **AWS Outposts**, partnering with **25+** engineering/manufacturing teams to enable factory-level provisioning before shipment.
– Removed post-delivery activation delays by shifting provisioning left, accelerating time-to-productivity and contributing to approximately **$156M** in monthly AWS revenue impact.
– Built an **AI** support assistant using Amazon **Bedrock** and **Model Context Protocol** (MCP) that analyzes docs, logs, and APIs to auto-resolve and route tickets.
– Hardened **LLM** workflows with allowlisted tool calls, scoped credentials, redaction, materially improving on-call.
– Developed "Live Update Replay" for **EC2**, enabling real-time validation, controlled rollout across millions of servers.
– Stream-processed rollout telemetry using **Flink** to detect anomalies early (error-rate/latency drift), gate rollout.
– Delivered observability with **Prometheus, Grafana** (integrated with **CloudWatch**), reducing MTTD by **40%.**
– Implemented API pagination, filtering, RBAC-aware views in the **React UI**, improving usability for large orgs.
– Optimized **DynamoDB** access pattern by redesigning partition/sort key, adding targeted **GSIs** for admin-console queries, reducing p95 read latency by **45%,** eliminating hot partitions under bursty enterprise onboarding.
– Designed secure multi-account authorization using **AWS Organization, IAM, STS**, enforcing least-privilege.

Software Developer 2, Dell Technologies, India — Jan 2019 - Jul 2022

– Owned end-to-end delivery of key modules of an internal analytics platform centralizing **sales, inventory, manufacturing, operations** data using **Java, Kafka, Flink, React, K8** while working in **Agile** environment.
– Led migration from monolithic analytics backend to **microservices** by carving out domain-aligned services (sales, inventory, manufacturing), introducing contract-first **REST** APIs and async integration via **Kafka**.
– Developed executive dashboards using **React, TypeScript** with reusable table/chart components for regional sales, product performance, stock levels, and forecast vs actual views.
– Tuned **OracleDB** reporting stores by designing composite indexes, partitioning large fact tables, and rewriting heavy joins into pre-aggregated rollups, improving p95 query times by **50%+.**
– Integrated **ERP**, sales databases, inventory systems, manufacturing logs into a unified analytics pipeline with consistent schemas and reconciliation checks.
– Built event-driven ingestion using **Kafka** for reliable change capture, replay, decoupled processing across domains.
– Implemented near-real-time KPI computation with **Flink** streaming jobs, batch backfills for late/duplicate events.
– Owned end-to-end delivery from schema design and ingestion to APIs and **React UI** performance (pagination, memoization, virtualization), increasing adoption across distributed teams.
– Introduced **Redis** caching for hot aggregate, dimension lookup; lowered latency, load on core databases.
– Established monitoring using **Prometheus, Grafana** for service health, latency, pipeline freshness, consumer lag.

EDUCATION

The University of Texas at Dallas, TX - Master of Science in Computer Science
Madanapalle Institute of Technology and Science, India - Bachelor's in Computer Science & Engineering

TECHNICAL SKILLS

Programming Languages : Java, TypeScript, Python, C#, JavaScript, SQL, Bash, Go, HTML, Kotlin, Scala
Frameworks/Technologies : Spring Boot, React, Node.js, Kafka, Flink, Spark, gRPC, RESTful APIs, GraphQL, Docker, Kubernetes, Helm, Terraform, Prometheus, Grafana, Amazon Redshift, Redis, Git, Linux
Databases/Cloud : AWS, Azure, GCP, Oracle Cloud, PostgreSQL, MySQL, DynamoDB, Amazon Aurora, MongoDB